

Derivatives and Fixed Income

PGE M1 Fall 2026

Jason Kang

Session 8: Put-Call Parity



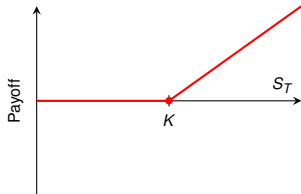
About me

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- ▷ jason.kang@skema.edu

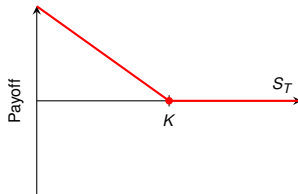
Payoff Diagrams: Calls and Puts



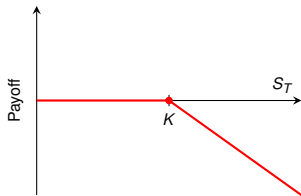
Long Call



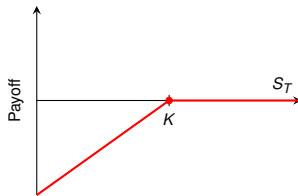
Long Put



Short Call



Short Put



Payoff Tables



	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Put	$-P$	$K - S_T$	0
Short Put	$+P$	$S_T - K$	0
Long Call	$-C$	0	$S_T - K$
Short Call	$+C$	0	$K - S_T$

Note, for a long call position, a payoff of $\max\{S_T - K, 0\}$ is equivalent to a payoff of $S_T - K$ when $S_T > K$ and 0 otherwise, as the option is not exercised.

Payoff Tables



	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Stock	$-S_0$	S_T	S_T
Short Stock	$+S_0$	$-S_T$	$-S_T$
Borrow (sell bond)	$+S_0$	$-S_0(1+r)^T$	$-S_0(1+r)^T$
Lend (buy bond)	$-S_0$	$S_0(1+r)^T$	$S_0(1+r)^T$
Borrow (sell bond)	$+\frac{K}{(1+r)^T}$	$-K$	$-K$
Lend (buy bond)	$-\frac{K}{(1+r)^T}$	K	K

Technically, you can borrow and lend any amount today but S_0 and $\frac{K}{1+r}$ are the ones relevant.

Protective Put



Suppose you currently own Apple stock, which you bought today at $S_0 = 323.80952$. To hedge a potential fall in prices, you enter a long put position at strike $K = 350$ with an expiration date of 1 year at price $p = 23.81$. Assume $r = 5\%$.

- a. Draw the payoff diagram
- b. Draw the profit diagram

Protective Put



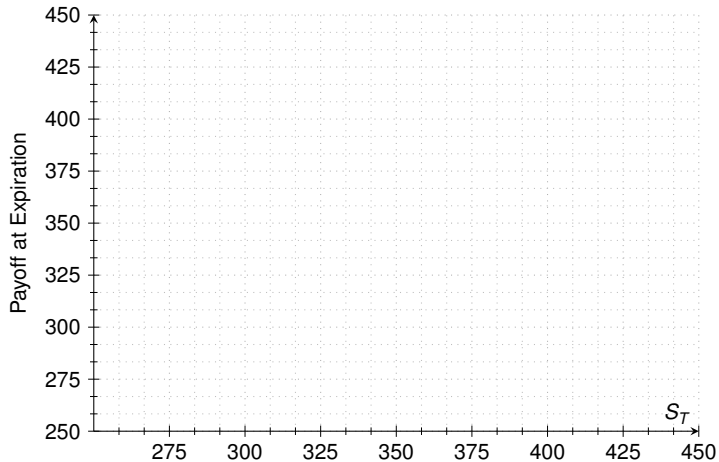
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	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Stock			
Long Put			
Total Payoff			

Protective Put



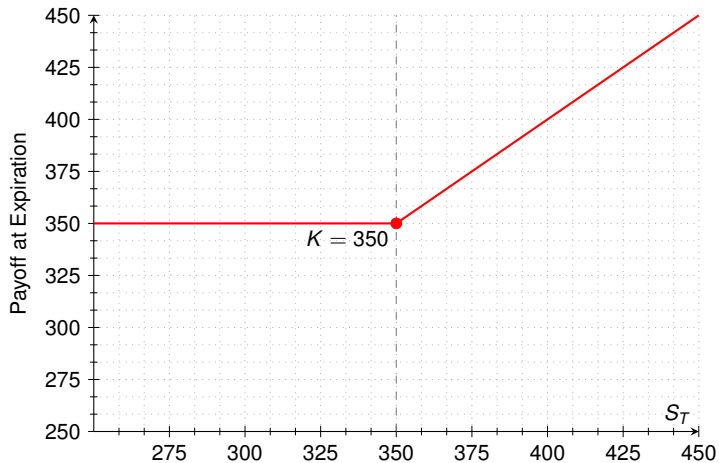
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Protective Put



a. Draw the payoff diagram.



Protective Put



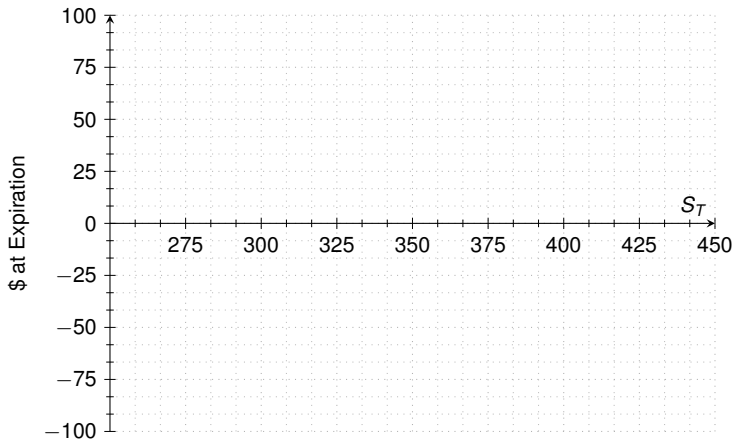
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	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Stock	-323.80952	S_T	S_T
Long Put	-23.81	$350 - S_T$	0
Total Payoff	-347.61952	350	S_T

Protective Put



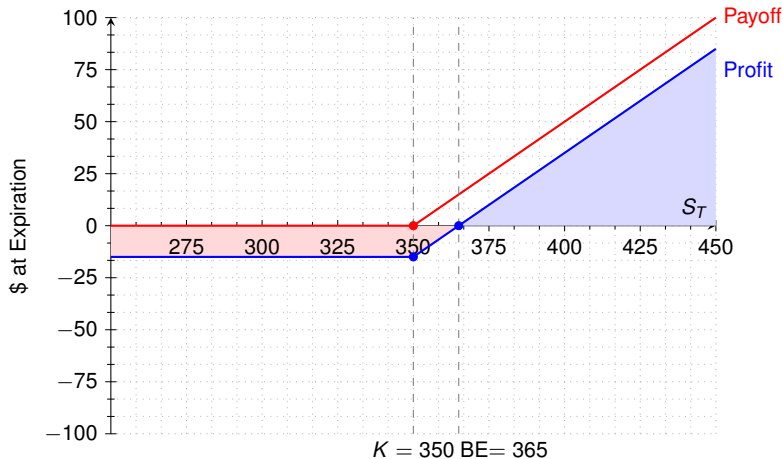
b. Draw the profit diagram. Assume you paid a premium of \$23.81 for the put ($23.81 \times 1.05 = 25$), and note $S_0 \times 1.05 = 323.81 \times 1.05 = 340$.



Protective Put



b. Draw the profit diagram. Assume you paid a premium of \$23.81 for the put ($23.81 \times 1.05 = 25$), and note $S_0 \times 1.05 = 323.81 \times 1.05 = 340$.



Replicating Portfolio



Suppose you lend \$333.33 for one year at an annual interest rate of $r = 5\%$. At the same time, you enter a long call position to buy Apple stock at strike $K = 350$ with a maturity of 1 year at price $c = 14.2867$.

- a. Draw the payoff diagram
- b. Draw the profit diagram

Replicating Portfolio



Suppose you lend \$333.33 for one year at an annual interest rate of $r = 5\%$. At the same time, you enter a long call position to buy Apple stock at strike $K = 350$ with a maturity of 1 year at price $c = 14.2867$.

- a. Draw the payoff diagram
- b. Draw the profit diagram

Replicating Portfolio



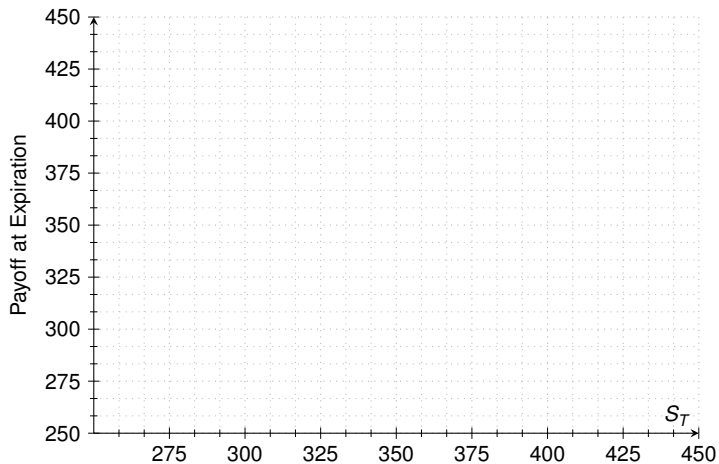
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	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Lend (Bond)			
Long Call			
Total Payoff			

Replicating Portfolio



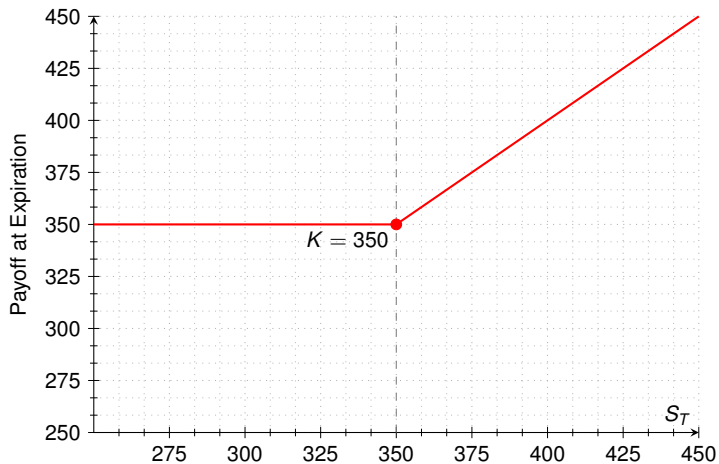
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Replicating Portfolio



a. Draw the payoff diagram.



Replicating Portfolio



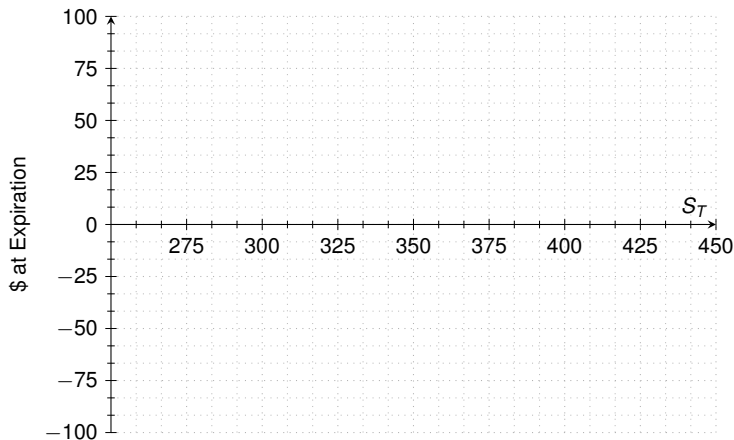
Suppose you lend \$333.33 for one year at an annual interest rate of $r = 5\%$. At the same time, you enter a long call position to buy Apple stock at strike $K = 350$ with a maturity of 1 year at price $c = 14.2867$.

	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Lend (Bond)	-333.33	350	350
Long Call	-14.2867	0	$S_T - 350$
Total Payoff	-347.6167	350	S_T

Replicating Portfolio



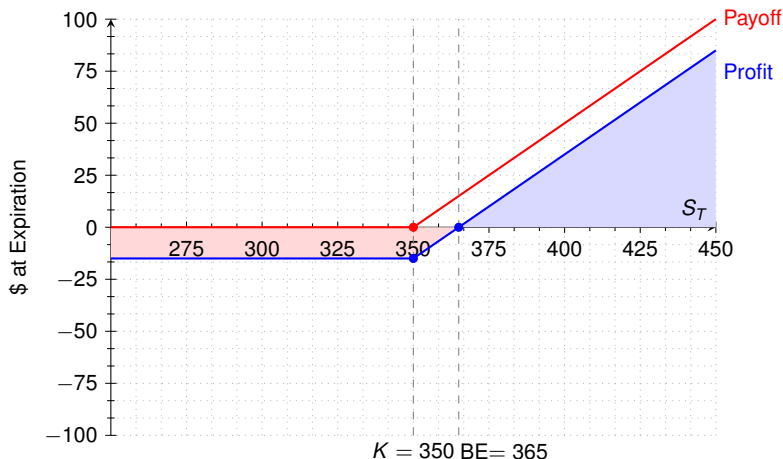
b. Draw the profit diagram. Assume you paid a premium of \$14.2867 for the call ($14.2867 \times 1.05 = 15$), and note the loan matures to $333.33 \times 1.05 = 350$.



Replicating Portfolio



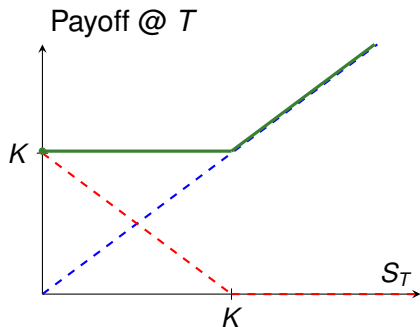
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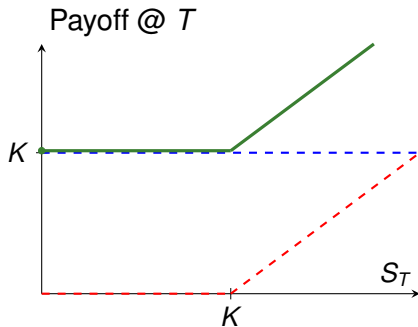
Put-Call Parity



Two portfolios give you the same payoff



Protective put = Stock + Put



Replicating P. = R.F. Bond + Call

They must have the same price today!!

Put-Call Parity



By the law of one price (LOOP) and no-arbitrage, we must have

$$p + S_0 = \frac{K}{(1+r)^T} + c$$

where

- ▷ p : price of a European put
- ▷ S_0 : price of the underlying at date 0
- ▷ K : the strike price
- ▷ T : expiration date
- ▷ c : price of a European call



Put-Call Parity

By the law of one price (LOOP) and no-arbitrage, we must have

$$p + S_0 = \frac{K}{(1+r)^T} + c$$

Assumptions.

- ▷ Both the put and call are written on the same K and T
- ▷ Both are European options
- ▷ The underlying is a non-dividend paying stock
- ▷ There are no transaction costs
- ▷ Everyone can borrow and lend at the risk-free interest rate, and is subject to the same tax treatment

Put-Call Parity



From the previous examples, we had

$$23.81 + 323.81 = 347.62 = 333.33 + 14.29$$

What if PCP doesn't hold?

Recall: Payoff Tables



	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Put	$-P$	$K - S_T$	0
Short Put	$+P$	$S_T - K$	0
Long Call	$-C$	0	$S_T - K$
Short Call	$+C$	0	$K - S_T$

▷ And

	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Stock	$-S_0$	S_T	S_T
Short Stock	$+S_0$	$-S_T$	$-S_T$
Borrow (sell bond)	$+S_0$	$-S_0(1+r)^T$	$-S_0(1+r)^T$
Lend (buy bond)	$-S_0$	$S_0(1+r)^T$	$S_0(1+r)^T$
Borrow (sell bond)	$+\frac{K}{(1+r)^T}$	$-K$	$-K$
Lend (buy bond)	$-\frac{K}{(1+r)^T}$	K	K

Arbitrage



Find an arbitrage strategy. Assume the underlying is a non-dividend paying stock, and that the prices of put and call options below are over the same strike price and are European.

a. $P = 5, C = 3, S_0 = 8, K = 8, r = 0\%$

Step 1: plug into put-call-parity

Step 2: Identify which of the options is “cheap”, and which is “expensive”.

Arbitrage



a. $P = 5, C = 3, S_0 = 8, K = 8, r = 0\%$

Step 3: Make a payoff table

	Today	Payoff at T	
		$S_T < K$	$S_T > K$
<u>1</u>			
		<u>2</u>	
<u>4</u>			
Payoff	<u>5</u>	<u>3</u>	

1. Write down the initial trades.
2. Cancel out S_T on the payoff table, then retroactively identify what action you must take.
3. Set the balance to 0
4. Describe the action to set the balance to 0 (it's either borrow or lend). If the riskless rate $\neq 0$, make sure to adjust for the time value of money.
5. Calculate payoffs.

Arbitrage



a. $P = 5, C = 3, S_0 = 8, K = 8, r = 0\%$

	Payoff at T	
	$S_T < K$	$S_T > K$
Today		
Payoff		

Arbitrage



a. $P = 5, C = 3, S_0 = 8, K = 8, r = 0\%$

- This time, set the balance to 0 today instead of at date T .

Today		Payoff at T	
		$S_T < K$	$S_T > K$
Payoff	<u>3</u> 0		

Arbitrage



b. $P = 3, C = 2, S_0 = 10, K = 9, r = 0\%$

	Payoff at T	
	$S_T < K$	$S_T > K$
Today		
Payoff		

Arbitrage



c. $P = 3, C = 4, S_0 = 11, K = 12, r = 0\%$

	Payoff at T	
	$S_T < K$	$S_T > K$
Today		
Payoff		

Arbitrage



d. $P = 3, S_0 = 10, K = 15, r = 0\%$

	Payoff at T	
	$S_T < K$	$S_T > K$
Today		
Payoff		

Arbitrage



e. $C = 2, S_0 = 8, K = 5, r = 0\%$

	Payoff at T	
	$S_T < K$	$S_T > K$
Today		
Payoff		

Put-Call Parity



Thus, due to LOOP + no-arbitrage, we have that

$$p + S_0 = \frac{K}{(1+r)^T} + c$$

or in the continuous version,

$$p + S_0 = Ke^{-rT} + c$$

and with the present-value of dividends D ,

$$p + S_0 = \frac{K}{(1+r)^T} + c + D$$

Lower Bound on the value of c



The long call has the following payoff

	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Call	$-c$	0	$S_T - K$

Compare to a long stock + borrowing at $K/(1+r)^T$

	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Stock	$-S_0$	S_T	S_T
Borrow	$+K/(1+r)^T$	$-K$	$-K$
Payoff	$-(S_0 - K/(1+r)^T)$	$S_T - K$	$S_T - K$

Which payoff is more attractive?



Lower Bound on the value of c

Since $S_T - K < 0$ when $S_T < K$, the lower bound on the value of a European call is given by

$$c \geq S_0 - \frac{K}{(1+r)^T}$$

or in the continuous version,

$$c \geq S_0 - Ke^{-rT}$$

and if the underlying stock pays dividends with present value D ,

$$c \geq S_0 - D - \frac{K}{(1+r)^T}$$

Lower Bound on the value of p



The long put has the following payoff

	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Long Put	$-p$	$K - S_T$	0

Compare to lending at $K/(1+r)^T$ + short stock

	Today	Payoff at T	
		$S_T < K$	$S_T > K$
Lend	$-K/(1+r)^T$	K	K
Short Stock	$+S_0$	$-S_T$	$-S_T$
Payoff	$S_0 - K/(1+r)^T$	$K - S_T$	$K - S_T$

Which payoff is more attractive?



Lower Bound on the value of p

Since $K - S_T < 0$ when $S_T > K$, the lower bound on the value of a European put is given by

$$p \geq \frac{K}{(1+r)^T} - S_0$$

or in the continuous version,

$$p \geq Ke^{-rT} - S_0$$

and if the underlying stock pays dividends with present value D

$$p \geq \frac{K}{(1+r)^T} + D - S_0$$

Exercise 11.1



What is a lower bound for the price of a 4-month call option on a non-dividend-paying stock when the stock price is \$28, the strike price is \$25, and the (continuously compounded) risk-free interest rate is 8% per annum?

3.66

Exercise 11.6



The price of a non-dividend-paying stock is \$19 and the price of a 3-month European call option on the stock with a strike price of \$20 is \$1. The (continuously compounded) risk-free rate is 4% per annum. What is the price of a 3-month European put option with a strike price of \$20?

$$p = 1.80$$



Exercise 11.10



A 4-month European call option on a dividend-paying stock is currently selling for \$5. The stock price is \$64, the strike price is \$60, and a dividend of \$0.80 is expected in 1 month. The (continuously compounded) risk-free interest rate is 12% per annum for all maturities. What opportunities are there for an arbitrageur?

Hint: Take the PV of D and add to the amount you borrow/lend today. Total profit = 0.56

